

How A Run Works Sample

Reading `json_keys_before` and `json_keys_after` in prompt evolution reports

Using your latest report at

[`prompt_evolution_report.csv`](/Users/oluwolejaiyeoba/Documents/GitHub/kv_cache_offloading/agentbench/prompt_evolution_report.csv) the ``json_keys_before`` and ``json_keys_after`` columns are showing the `**shape` of the data object`**` before and after each stage, using keys only.

Here's what each row is saying.

``task_input``

- ``json_keys_before` = none``
- ``json_keys_after` = {"repo", "base_commit", "problem_statement", "requirements", "selected_test_files_to_run", "works"}`

Meaning:

- before this stage, there is no structured task object yet
- after this stage, AgentBench has loaded the raw SWE-bench task fields into a task-shaped object

So this is the moment where the run gets its initial structured input.

``formatted_prompt``

- ``json_keys_before` = {task fields...}``
- ``json_keys_after` = {"prompt": "..."}``

Meaning:

- before this stage, the data is still spread across many separate task fields
- after this stage, those fields have been collapsed into one single prompt string

So this row is showing a big structural simplification:

- many dataset fields
- become one formatted prompt

``final_model_request``

- ``json_keys_before` = {"prompt": "..."}``
- ``json_keys_after` = {"model", "messages", "request_context", "agent_hints", "tool_choice"}``

Meaning:

- before this stage, we only have a plain prompt string
- after this stage, that prompt is wrapped into a real chat request object

This is one of the most important transformations:

- plain text prompt
- becomes OpenAI-style request structure

Especially:

- ``prompt`` turns into ``messages[0].content``
- request metadata is added
- routing hints are added

``system_context``

- ``json_keys_before`` = request object with `model/messages/request_context/agent_hints/tool_choice``
- ``json_keys_after`` = same thing + `"system_prompt"`

Meaning:

- before this stage, the request already exists
- after this stage, Deep Agents' system instruction layer is added

So structurally, this is a smaller change:

- the request stays a request
- but now it has system-level behavior instructions too

``tool_runtime_context``

- ``json_keys_before`` = request + `system_prompt``
- ``json_keys_after`` = same thing + `expected tools + parser fields``

Meaning:

- before this stage, the request has prompt and system context
- after this stage, it becomes explicitly tool-capable

New structure appears for:

- ``expected_builtin_tools``
- ``tool_parser_names_seen``
- ``tool_parser_observed``

So this row is showing:

- the request is now enriched with tool/runtime capability context

``runtime_preprocessing``

- ``json_keys_before`` = tool-capable request
- ``json_keys_after`` = same thing + `prompt/cache metrics``

Meaning:

- before this stage, the request is already tool-capable
- after this stage, runtime preprocessing measurements are attached

New structure appears for:

- `prompt\_tokens`
- `cached\_prompt\_tokens`
- `cached\_input\_tokens`

So this is not a semantic prompt rewrite. It is a runtime-measurement enrichment step.

`model\_behavior`

- `json\_keys\_before` = runtime-prepared request state`
- `json\_keys\_after` = response/transcript/outcome state`

Meaning:

- before this stage, the object still represents the prepared request
- after this stage, the object represents what actually happened

This is the other biggest structural shift in the report:

- request-oriented structure
- becomes response/outcome-oriented structure

Now the keys are about:

- `messages`
- `tool\_calls`
- `observed\_tool\_call\_names`
- `observed\_tool\_result\_names`
- `finish\_reason`
- `response\_text`
- `workspace\_changed`

So this row is showing the transition from:

- what we were about to send / had prepared

to:

- what the model actually did and what outcome we got

The Big Picture

Across all rows, the JSON-shape story is:

1. no structure
2. raw task object
3. single prompt string
4. full model request object
5. request plus system instructions
6. request plus tool/runtime context
7. request plus runtime preprocessing metrics
8. final response/outcome object

So the `json\_keys\_before` / `json\_keys\_after` columns are basically showing the
object-shape evolution of the run from raw task input all the way to final model behavior.